Saurabh Surashe

MTS III at Mavenir Systems

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Professional Summary

Embedded Software Engineer with **4+ years of experience** developing production-grade software in **C/C++, Embedded Linux, multithreading, IPC, networking, and performance-critical systems**. Experienced in designing and optimizing concurrent software, real-time schedulers, protocol stacks, low-latency communication mechanisms, and system-level software for complex embedded platforms. Delivered **130+ production features and fixes** spanning software architecture, performance optimization, debugging, reliability, and customer deployments.

Strong interest in **embedded platforms, connectivity software, networking, semiconductor software, systems programming, and high-performance C/C++**.

Core Competencies

Programming: C, C++17, STL, Python, Java
Systems & Linux: Embedded Linux, pthreads, Multithreading, IPC, Shared Memory, Process/Thread Synchronization, GDB, Intel VTune
Networking: TCP/IP, UDP, Socket Programming, Wireshark, Protocol Analysis
Wireless & Protocols: Wi-Fi, NB-IoT, MAC, RRC, S1AP, ASN.1, 3GPP Protocols
Software Engineering: Concurrent Programming, Event-Driven Programming, Real-Time Systems, Low-Latency Systems, Performance Optimization, Distributed Systems, Debugging, Root-Cause Analysis
AI–ML: Sentence Transformers, Semantic Search, Embeddings, TensorFlow, Keras

Professional Experience

Mavenir Systems Pvt. Ltd. July 2022 – Present
Member of Technical Staff I → Member of Technical Staff III *Bengaluru, India*

- Develop production-grade **C/C++ software on Embedded Linux** for real-time communication systems, contributing to software architecture, scheduler implementation, protocol processing, performance optimization, debugging, and production delivery.
- Redesigned latency-sensitive **C scheduler algorithms**, improving uplink throughput by approximately **40%** while maintaining deterministic execution requirements for real-time embedded workloads.
- Delivered more than **130 production features, enhancements, and fixes** spanning embedded software functionality, protocol processing, runtime optimization, performance improvements, and production reliability.
- Designed and implemented **shared-memory IPC** mechanisms for low-latency communication between software modules, addressing synchronization, data sharing, and runtime efficiency requirements.
- Developed concurrent and event-driven software using **pthreads, synchronization primitives, IPC mechanisms, and Linux system programming** for performance-sensitive workloads.
- Developed and maintained more than **60 production KPIs** for runtime diagnostics, system observability, scheduler monitoring, performance analysis, and software health monitoring.
- Optimized CPU-intensive execution paths using **Intel VTune and GDB**, identifying performance bottlenecks and reducing execution latency in critical software paths.
- Performed system-level root-cause analysis using **logs, memory analysis, protocol traces, Wireshark, GDB, and low-level debugging**, resolving complex production issues involving performance, synchronization, memory, and protocol behavior.
- Developed protocol software involving **MAC, RRC, S1AP, ASN.1 encoding/decoding, scheduling, and network communication**, integrating software across multiple system components.
- Implemented software enhancements for non-terrestrial communication systems involving **timing management, scheduling, and dynamic satellite-related parameters**.
- Collaborated with Layer-1, Platform, QA, DevOps, and customer engineering teams across software design, implementation, validation, deployment, and production support.
- Mentored junior engineers through technical discussions, debugging sessions, code reviews, and software design guidance.

Projects

AI-Powered Job Discovery & Resume Matching Platform | *Python, SQL, Sentence Transformers*

- Developed a modular Python platform for automated job discovery and semantic resume-to-job matching using the `all-MiniLM-L6-v2` transformer model and embedding-based similarity scoring.
- Built asynchronous scraping pipelines using `asyncio` to concurrently fetch and process job opportunities from Workday, Greenhouse, Eightfold, and SmartRecruiters ATS platforms.
- Implemented vector-based semantic search using NumPy to efficiently calculate resume-to-job similarity scores without database-side vector extensions.
- Designed REST APIs using FastAPI for resume parsing, job indexing, candidate profiles, matching configuration, and automated job processing.
- Built automated notification workflows delivering HTML job digests through SMTP and deployed scheduled processing using cloud/serverless pipelines.

Achievements

Mavenir Spotlight Award – Received multiple times for high-impact production features, technical contributions, and successful customer software releases.

Co-authored **"Enhancing Cyber Security: Malicious Web Page Detection using Machine and Deep Learning"**, AINA 2025 (*SpringerLink*)

Co-authored **"Privacy and Security Comparison of Web Browsers: A Review"**, AINA 2022 (*SpringerLink*)

Education

National Institute of Technology Surathkal — Karnataka
Master of Technology in Computational Data Science — *June 2020 – May 2022*

MIT College of Engineering Pune — Maharashtra
Bachelor of Engineering in Mechanical Engineering — *June 2015 – May 2019*